BSAVA Information Architecture, Data Orchestration & Historic Migration Strategy

Executive Workshop Strategy Document & Technical Guide

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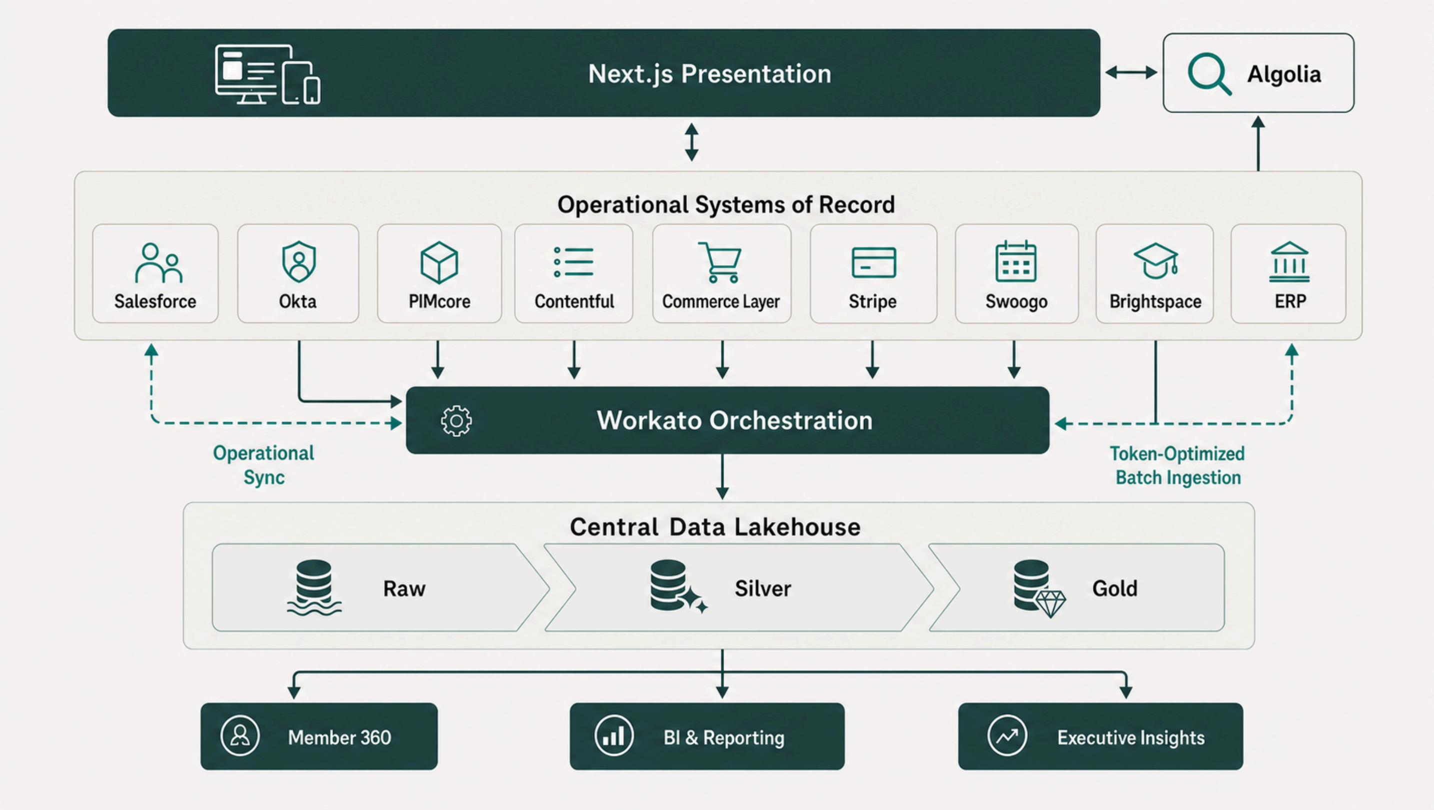
Executive Summary & Strategic Context

The British Small Animal Veterinary Association (BSAVA) is executing a comprehensive digital transformation, transitioning from legacy, monolithic IT systems to a modern, composable **MACH (Microservices, API-first, Cloud-native, Headless)** architecture.

While the headless presentation layer (Next.js), content management (Contentful), product catalog (PIMcore), search engine (Algolia), and payment processing (Stripe) provide a high-performance frontend experience, the true enterprise value of this transformation relies on a robust **Information Architecture (IA)**, clear **Systems of Record (SoR)**, seamless **Data Orchestration (Workato)**, a scalable **Central Data Lakehouse**, and a disciplined **Historic Data Migration Strategy**.



Key Goals of this Strategy Document:

1. **Define Systems of Record (SoR)**: Establish clear ownership boundaries for every data domain across BSAVA's headless platform to eliminate data duplication, operational latency, and conflicting member states.
2. **Design Workato Data Orchestration**: Specify how Workato acts as the single integration hub—not only for operational system-to-system syncs, but also for streaming clean transactional, behavioral, and log data into the central analytical layer.
3. **Compare Data Lakehouse Storage Platforms**: Perform a detailed, objective comparative analysis of **Snowflake**, **Microsoft Fabric**, and **Databricks** across functionality, architecture, pricing mechanics (Credits vs. Capacity Units vs. DBUs), and Total Cost of Ownership (TCO) tailored to BSAVA.
4. **Deliver a Phased Historic Migration Strategy**: Provide an end-to-end framework to extract, cleanse, and load historic legacy data into both live operational SoRs and the central Data Lakehouse without disrupting active business operations.

1. Information Architecture (IA) & Systems of Record (SoR) Matrix

In a composable architecture, no single operational application acts as a monolithic database. Instead, specialized best-of-breed platforms own specific data domains. To guarantee data integrity and compliance (including UK GDPR), BSAVA strictly enforces the **Single Source of Truth (SoR)** principle: *Every piece of data must have exactly one authoritative operational owner.*

1.1 Data Domain & System of Record Ownership Matrix

Data Domain / Entity	System of Record (SoR)	Primary Key / Identifier	Key Attributes Owned	Downstream Consuming Systems
Person / Contact Master	**Salesforce (CRM)**	`Contact ID` (`003...`)	Demographics, Contact Details, Committee Roles, CRM Interaction History	CIAM (Okta), Commerce Layer, Swoogo, Brightspace, Data Lakehouse
Credentialed User Identity	**Okta / Auth0 (CIAM)**	`User UID` (`usr_...`)	Password Hashes, MFA Tokens, SSO Sessions, Edge JWT Claims, Security Logs	Next.js Frontend, Brightspace (SSO), Swoogo (SSO)
Product Catalog & Entitlements	**PIMcore (PIM / DAM)**	`SKU / Product ID`	Book Catalog, Membership SKUs, Digital Assets, Entitlement Definitions & Access Rules, Pricing Tiers, Inventory	Next.js Frontend, Algolia, Commerce Layer, Salesforce, Workato
Editorial & CMS Content	**Contentful (CMS)**	`Content Entry ID`	News Articles, Clinical Guidelines, Landing Pages, Editorial Metadata	Next.js Frontend, Algolia
Carts, Orders & Subscriptions	**Commerce Layer (OMS)**	`Order ID` (`ord_...`) / `Sub ID`	Active Cart State, Membership Tiers & Subscription States, Order Line Items, Discounts, Delivery Status, Order History	Salesforce, Stripe, PIMcore, Workato, Accounting ERP, Data Lakehouse
Payment Gateway Data	**Stripe**	`PaymentIntent ID` (`pi_...`)	Credit Card Tokens, Payment Intents, Charges, Refunds, Gateway Payout Logs	Commerce Layer, Accounting ERP, Data Lakehouse

Data Domain / Entity	System of Record (SoR)	Primary Key / Identifier	Key Attributes Owned	Downstream Consuming Systems
Events & Conferences	**Swoogo**	`Attendee ID` / `Event ID`	Event Schedules, Ticket Allocations, Delegate Rosters, Speaker Profiles	PIMcore (via Workato), Salesforce, Data Lakehouse
Learning Management & CPD	**Brightspace (D2L LMS)**	`Student ID` / `Course ID`	Course Catalog, Module Enrolments, Progress %, Quiz Scores, CPD Credit Hours	PIMcore (via Workato), Salesforce, Algolia, Data Lakehouse
Financial Ledger & Invoices	**Accounting (ERP)**	`Invoice ID` / `Journal ID`	General Ledger, Chart of Accounts, Formal Invoices, Tax Compliance, Payout Rec	Data Lakehouse, Executive Reporting
Search Operational Cache	**Algolia**	`ObjectID`	Search Indexes, Facets, Filter Attributes, Ranking Rules *(Cache only)*	Next.js Frontend
Enterprise Historical Analytics	**Data Lakehouse**	Unified `Member_360_ID`	Historical Member Telemetry, Cross-Domain BI, Trend Analysis, Executive KPIs	Power BI / Tableau, Analytics Dashboards

1.2 Deep-Dive Operational System Analysis

1. Salesforce (CRM) — Core Customer & Contact History SoR

- **Role**: Serves as the authoritative source of truth for member contact profiles, committee roles, and CRM interaction history. Note: **Membership Tiers** are managed in **Commerce Layer**, and **Entitlement Definitions & Access Rules** are governed in **PIMcore**.
- **Data Format & Access**: Relational Salesforce Objects (`Contact` , `Account` , `Committee_Role__c`). Accessible via REST, SOAP, and Bulk API 2.0.
- **Native Reporting Capabilities**: Strong operational reports and dashboards for active sales pipelines and contact lists.
- **Analytical Limitations**: Limited in handling high-frequency behavioral telemetry (e.g., website clickstreams, course playback progress) and multi-year cross-domain trend queries due to storage governor limits and API costs.

2. Okta / Auth0 (CIAM) — Identity & Session Governance

- **Role**: Master repository for user credentials, authentication security rules, and edge JWT token issuance.
- **Data Format & Access**: JSON directory objects accessible via System Log API and Management REST API.
- **Reporting & Analytics Role**: Provides security audits and login telemetry. Authentication logs are streamed to the Data Lakehouse for security monitoring and login behavior analysis.

3. PIMcore (PIM / DAM) — Structured Catalog, Asset & Entitlement SoR

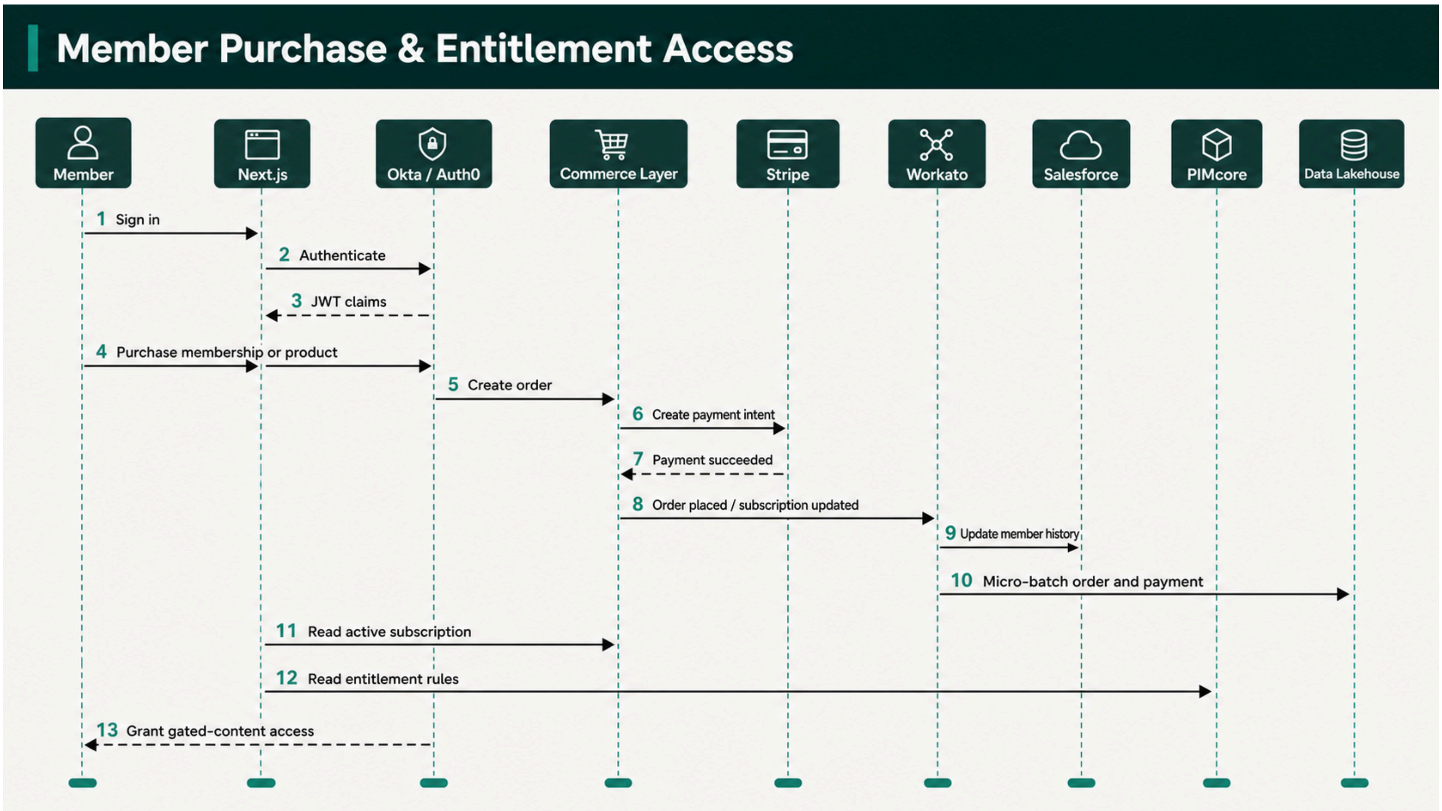
- **Role**: Single source of truth for all sellable products (BSAVA manuals, digital publications, veterinary guidelines), membership packages, and **Entitlement Definitions & Access Rules** (defining which digital assets, clinical publications, or gated resources each product or tier unlocks).
- **Data Format & Access**: Structured relational/document objects accessible via GraphQL and REST APIs.
- **Reporting & Analytics Role**: Master product metadata and entitlement definitions are snapshotted to the Data Lakehouse to enrich transaction and access reports.

4. Contentful (CMS) — Marketing & Editorial Content Master

- **Role**: Authoritative store for un-gated marketing copy, news updates, clinical articles, and media content models.
- **Data Format & Access**: JSON document models accessible via Content Delivery API (CDA) and Content Management API (CMA).
- **Reporting & Analytics Role**: Content metadata is indexed to Algolia for instant site search and streamed to the Data Lakehouse to track content engagement performance.

5. Commerce Layer (OMS) & Stripe — Order, Membership Tier & Payment Processing

- **Role**: Commerce Layer manages **Membership Tiers**, active subscription states, order fulfillment workflows, and cart states. Stripe securely executes credit card authorizations, subscriptions, and payout settlements.
- **Data Format & Access**: Webhooks (`order.placed` , `subscription.updated` , `payment_intent.succeeded`) and REST APIs.
- **Reporting & Analytics Role**: Order headers, line items, subscription states, tax breakdowns, and payment gateway logs are ingested into the Data Lakehouse for daily financial reconciliation and sales analytics.



6. Swoogo — Events Management Platform

- **Role**: Manages congress and CPD event creation, ticket pricing, delegate registrations, and attendance rosters.
- **Data Format & Access**: REST API and Webhooks (`event.updated` , `registration.completed`).
- **Reporting & Analytics Role**: Event delegate profiles and attendance records are synced to Salesforce for contact history and streamed to the Data Lakehouse for event ROI analysis.

7. Brightspace (D2L LMS) — Learning & CPD Accreditation

- **Role**: Master platform for online veterinary courses, educational modules, learning progress, and CPD credit records.
- **Data Format & Access**: Data Hub (bulk CSV extractions via REST API) and real-time Event Notification Services (ENS).
- **Reporting & Analytics Role**: Course completions and earned CPD hours are synced to Salesforce to update member profiles and streamed to the Data Lakehouse to track educational engagement trends.

8. Accounting (ERP) — Financial Ledger & Compliance

- **Role**: Authoritative ledger for financial accounts, chart of accounts, tax compliance (VAT), and audit-ready invoicing.

- **Data Format & Access**: REST API / Scheduled File Export.
- **Reporting & Analytics Role**: General Ledger entries and monthly trial balances are ingested into the Data Lakehouse for executive revenue reporting and reconciliation against Stripe/Commerce Layer.

1.3 Data Residency & UK GDPR Compliance

To maintain strict compliance with **UK GDPR** and national data protection regulations, BSAVA enforces explicit geographic binding across all software components:

- **UK / EU Region Provisioning**: All operational Systems of Record (Salesforce, Commerce Layer, PIMcore) and the primary Data Lakehouse cluster must be provisioned exclusively within **UK South (Azure)** or **EU West / London (`eu-west-2` , AWS)** regions. Storage or transit of unmasked member data in US East or unapproved third-country cloud environments is strictly prohibited.
- **Vendor UK Availability**: All three evaluated Data Lakehouse platforms (Snowflake, Microsoft Fabric, Databricks) support fully certified, native UK-region deployments (Azure UK South and AWS London).
- **Steering Committee Approval**: The IT Steering Committee must formally sign off on the chosen cloud region configuration prior to initiating Phase 2 migration data ingestion.

1.4 Data Classification & PII Masking Policy

To operationalize data privacy and security, data entities across BSAVA's SoR matrix (Section 1.1) are categorized into four standard classification tiers, each governed by specific Lakehouse Silver/Gold zone masking rules:

Sensitivity Tier	Associated SoR Domains (Section 1.1)	Access & Encryption Control	Data Lakehouse Silver/Gold Zone Masking Policy
Public	Editorial Content (Contentful), Product Metadata & Entitlement Definitions (PIMcore), Public Event Schedules (Swoogo)	Open read access; standard TLS 1.3 in transit.	Ingested unmasked into Silver and Gold analytical zones.
Internal	General Ledger Accounts (ERP), Aggregate Course Analytics (Brightspace)	Role-based internal access (RBAC); encrypted at rest.	Available for general BI and executive reporting dashboards.
Restricted	Member Contacts (Salesforce), Membership Subscription Tiers (Commerce Layer), CPD Transcript Details (Brightspace)	Strictly enforced RBAC; encrypted at rest (AES-256).	**Pseudonymized**: Email and names masked in Silver zone; unmasked in Gold zone via role-gated policies.
Confidential	CIAM Passwords/Hashes (Okta), Payment Card Tokens (Stripe), Member Safeguarding Notes	Cryptographically isolated; strict least-privilege RBAC.	**Strictly Masked / Omitted**: Passwords/tokens never loaded into Lakehouse. DOB & sensitive notes SHA-256 hashed or tokenized.

Mandatory Field-Level Masking Rules:

1. **Payment Tokens & Financial Identifiers**: Stripe `PaymentIntent` IDs and payment method tokens are tokenized. Raw credit card numbers, CVVs, and banking details are never ingested into the Data Lakehouse.
2. **Date of Birth (DOB) & Contact Details**: Member DOBs are truncated to birth year in the Silver zone. Full email addresses are masked (e.g., `j***@domain.co.uk`) unless queried via an authorized Member 360 security role in the Gold datamart.

3. **Safeguarding & Sensitive Notes**: Free-text notes containing sensitive member correspondence or accessibility requirements are excluded from general lakehouse ingestion or processed via automated regex redactions.

2. Data Orchestration with Workato for Analytics & Reporting

Workato serves as the central **Integration Platform as a Service (IPaaS)** for BSAVA. To maintain clean architecture, Workato's responsibilities are divided into two distinct processing flows:

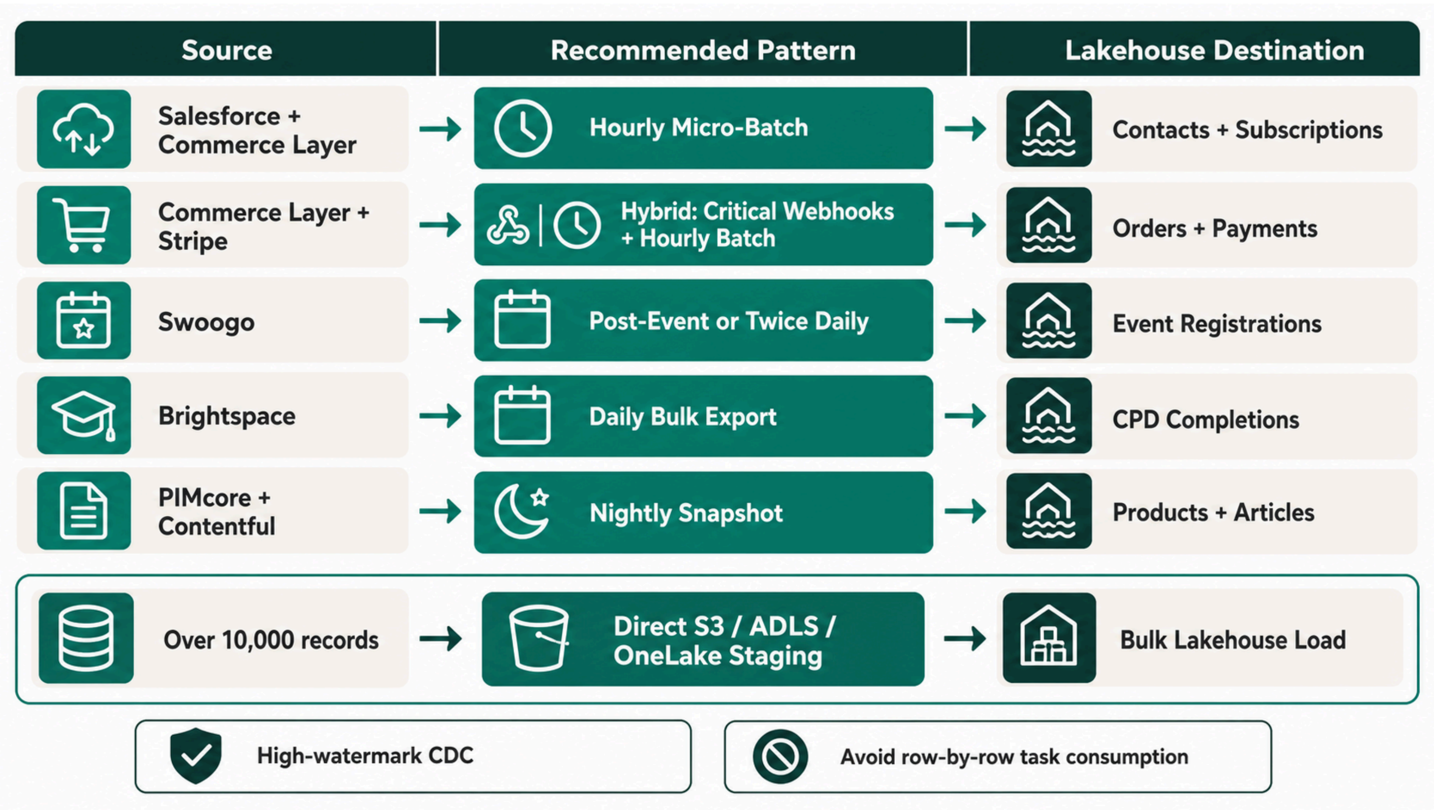
1. **Operational Orchestration (System-to-System Sync)**: Direct recipes that synchronize state between operational SoRs (e.g., updating Salesforce when an order is completed in Commerce Layer).
2. **Analytical Data Ingestion (ETL / ELT Pipelines)**: Extracting operational events, change logs, and daily snapshots from SoRs and delivering them into the **Central Data Lakehouse** for business intelligence and reporting.

Important

Commercial Context & Token Management Constraint
BSAVA operates under a contracted commercial limit of **1,000,000 Workato Tasks/Tokens per year**.

Real-time webhook-driven integrations consume **1 Workato task per event payload**. In high-volume scenarios (e.g., thousands of user interactions, course progress updates, order status changes, and catalog updates), streaming raw events via real-time webhooks can rapidly deplete BSAVA's annual token quota.

Therefore, **the adoption of real-time webhooks for analytical data ingestion is not yet finalized** and represents a primary decision topic for the upcoming workshop. This section details both the real-time webhook pattern and token-optimized micro-batching / bulk staging alternatives.



2.1 Ingestion Patterns: Real-Time Webhooks vs. Token-Optimized Micro-Batching

To balance data freshness against BSAVA's **1,000,000 Workato token budget**, three ingestion patterns are evaluated for the client workshop:

Ingestion Pattern	Trigger Mechanism	Latency	Annual Token Impact	Recommended Use Cases & Decision Status
Real-Time Streaming	Operational Webhooks (`HTTP POST`)	< 5 Seconds	**High Risk** (~300k–600k+ tasks/yr if unthrottled)	*Under Review*: Evaluate exclusively for critical operational flows (e.g. instant checkout payment confirmation or security events). **Not recommended for general BI reporting** due to token cost.
Scheduled Micro-Batching	High-watermark CDC polling (e.g., 30–60 min intervals)	15–60 Minutes	**Low to Moderate** (~17k–35k tasks/yr)	**Primary Recommendation**: Workato runs once per interval, extracting all changed records (`updated_at > last_run`) in a single batch array, using 1 task per batch run.
Daily Bulk Stage	Scheduled cron (02:00 UTC) with Direct Storage Copy	24 Hours	**Minimal** (< 1,000 tasks/yr)	**Recommended for Catalog & Bulk Logs**: Direct API extraction writing bulk CSV/Parquet files into cloud storage staging (S3/ADLS/OneLake stage).

2.2 Workato Analytical Recipe Architecture (Token-Optimized Options)

To ensure reporting needs are met while respecting the **1,000,000 Workato task budget**, each analytical recipe is structured with a token-optimized design option:

Recipe 1: Member Profile & Subscription Ingestion (Salesforce & Commerce Layer → Data Lakehouse)

- **Real-Time Option**: Salesforce CDC (`ChangeEvent`) and Commerce Layer webhooks triggering on every Contact or Subscription modification (High token usage).
- **Token-Optimized Option (Recommended)**: Scheduled hourly micro-batch querying Salesforce REST API for contacts modified in the last 60 minutes and Commerce Layer API for subscription tier changes. Aggregates modified profiles and active subscription states into a JSON batch payload (1 task per hour = 8,760 tasks/year).
- **Destination**: Ingests into `raw_salesforce_contacts` and `raw_commercelayer_subscriptions` .

Recipe 2: Financial Transaction & Order Log Ingestion (Commerce Layer + Stripe → Data Lakehouse)

- **Real-Time Option**: Individual `order.placed` and `charge.succeeded` webhooks.
- **Token-Optimized Option**: Hybrid model—real-time webhooks for high-priority operational order fulfillment, combined with micro-batched hourly order log extractions for analytics, ensuring data lake ingestion runs in bulk arrays.
- **Destination**: Ingests into `raw_commerce_orders` and `raw_stripe_payments` .

Recipe 3: Event Attendance & Delegate Analytics (Swoogo → Data Lakehouse)

- **Real-Time Option**: Webhook per delegate check-in during conferences (Risks token spikes during major congress events).
- **Token-Optimized Option (Recommended)**: Post-event bulk sync or twice-daily scheduled poll fetching delegate registration arrays after event sessions conclude.

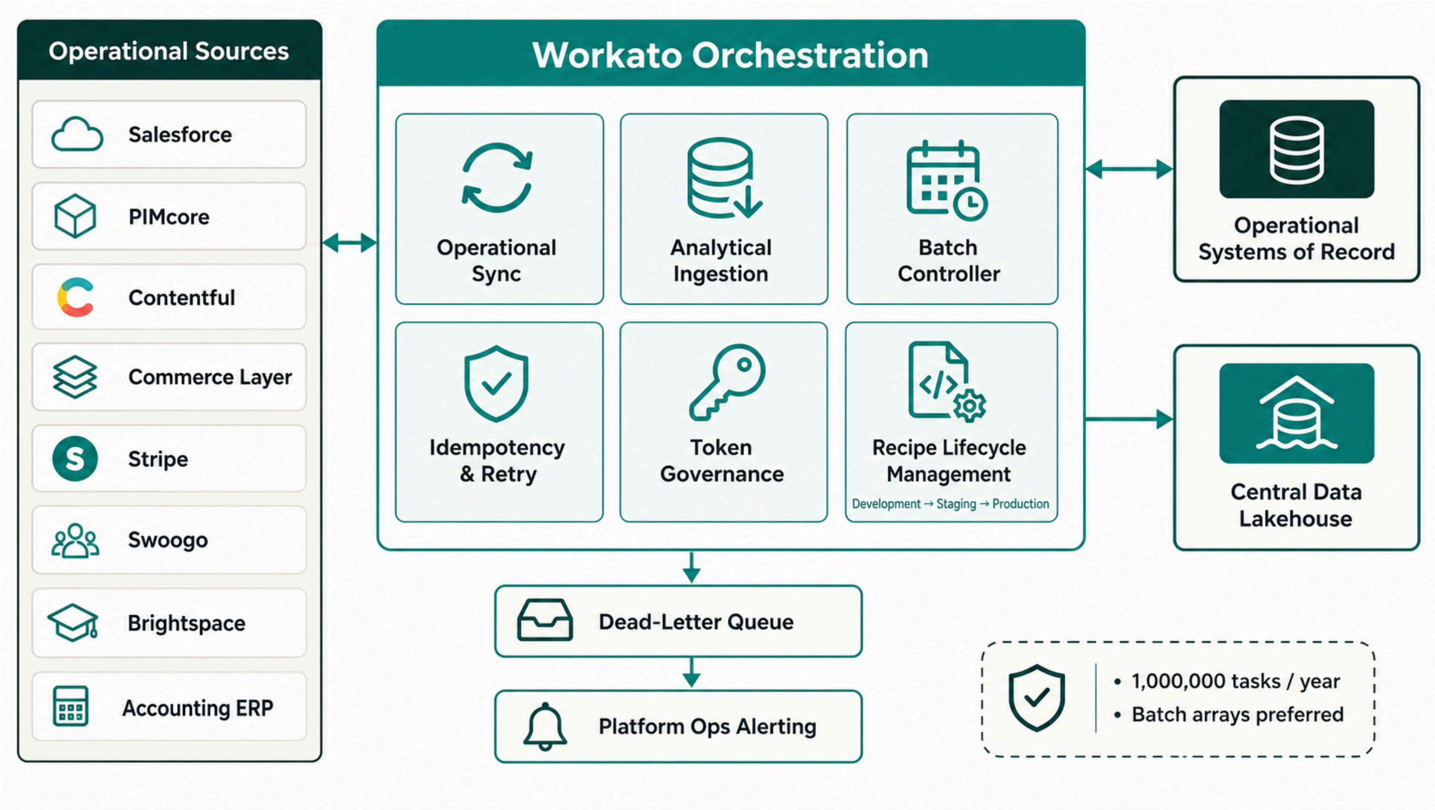
- **Destination**: Ingests into `raw_swoogo_registrations`.

Recipe 4: Learning & CPD Credit Ingestion (Brightspace → Data Lakehouse)

- **Real-Time Option**: Real-time event notifications on course progress.
- **Token-Optimized Option (Recommended)**: Automated daily pull from Brightspace **Data Hub** bulk CSV exports, landing complete course completion and CPD credit files into S3/ADLS for direct Lakehouse loading (365 tasks/year total).
- **Destination**: Ingests into `raw_brightspace_completions`.

Recipe 5: Product & Content Catalog Snapshots (PIMcore + Contentful → Data Lakehouse)

- **Strategy**: Nightly scheduled trigger at 01:00 UTC querying PIMcore GraphQL and Contentful API for active catalog metadata.
- **Destination**: Overwrites/appends daily snapshot tables `snap_pimcore_products` and `snap_contentful_articles`.



2.3 Workato Token Governance & Cost Control Policy

To safeguard BSAVA against unexpected Workato task overages, the integration architecture enforces four cost-governance rules:

1. **Batch Array Processing**:
   - Recipes must process collections of records as a single Workato batch payload rather than looping through individual records with `Repeat for each` steps (which consume 1 task per loop iteration).
2. **Direct Storage Staging (Bypassing IPaaS Row Processing)**:
   - For datasets larger than 10,000 records (e.g. historic migration data or full catalog exports), Workato triggers direct file transfer routines that push bulk GZIP/Parquet files directly into AWS S3 or Azure Blob Storage, allowing the Data Lakehouse to perform bulk native ingestion without consuming Workato row-processing tokens.
3. **Quota Monitoring & Automated Alerts**:
   - An automated Workato administrative recipe polls the Workato Usage API monthly. If task consumption exceeds **80,000 tasks/month** (approaching the 1/12th annual threshold of the 1,000,000 quota), an alert is issued to `#platform-ops` and the IT Steering Committee.

4. **Idempotency & Retry Governance**:
   - Transient API failures trigger exponential backoff retries capped at 3 attempts. Persistent errors payload land in a Dead-Letter Queue (DLQ) rather than spinning in retries that deplete token balances.
   - Workato includes the unique event signature in the Data Lakehouse insert statement. Target staging tables utilize deduplication keys (e.g., `QUALIFY ROW_NUMBER() OVER (PARTITION BY order_id ORDER BY ingested_at DESC) = 1`).
5. **Error Handling & Dead-Letter Queue (DLQ)**:
   - All Workato analytical recipes wrap network calls in `Try-Catch` blocks.
   - Transient failures (e.g., target database maintenance, API rate limits) automatically trigger **exponential backoff with jitter** (up to 5 retries over 30 minutes).
   - Unrecoverable failures payload are written to an AWS S3 / Azure Blob **Dead-Letter Queue (DLQ)** and trigger an automated notification to BSAVA's `#platform-ops` Slack channel.
6. **Environments & Recipe Lifecycle Management**:
   - Workato Recipe Lifecycle Management (LCM) enforces separate environments: `Development` → `Staging` → `Production`.
   - Hardcoded URLs or API credentials are strictly prohibited; environment properties and key vault secrets are injected dynamically.

3. Comparative Analysis: Data Lakehouse Platforms

To establish a central repository for historical reporting, member 360 analytics, and executive BI, BSAVA requires a modern cloud Data Lakehouse. We have evaluated the three leading industry platforms: **Snowflake**, **Microsoft Fabric**, and **Databricks**.

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│                          DATA LAKEHOUSE ARCHITECTURE LAYER                          │
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│  RAW LANDING ZONE (S3 / ADLS / OneLake)                                             │
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│  │ Raw JSON Events    │  │ Daily Change Logs  │  │ Historic Migration Files       │ │
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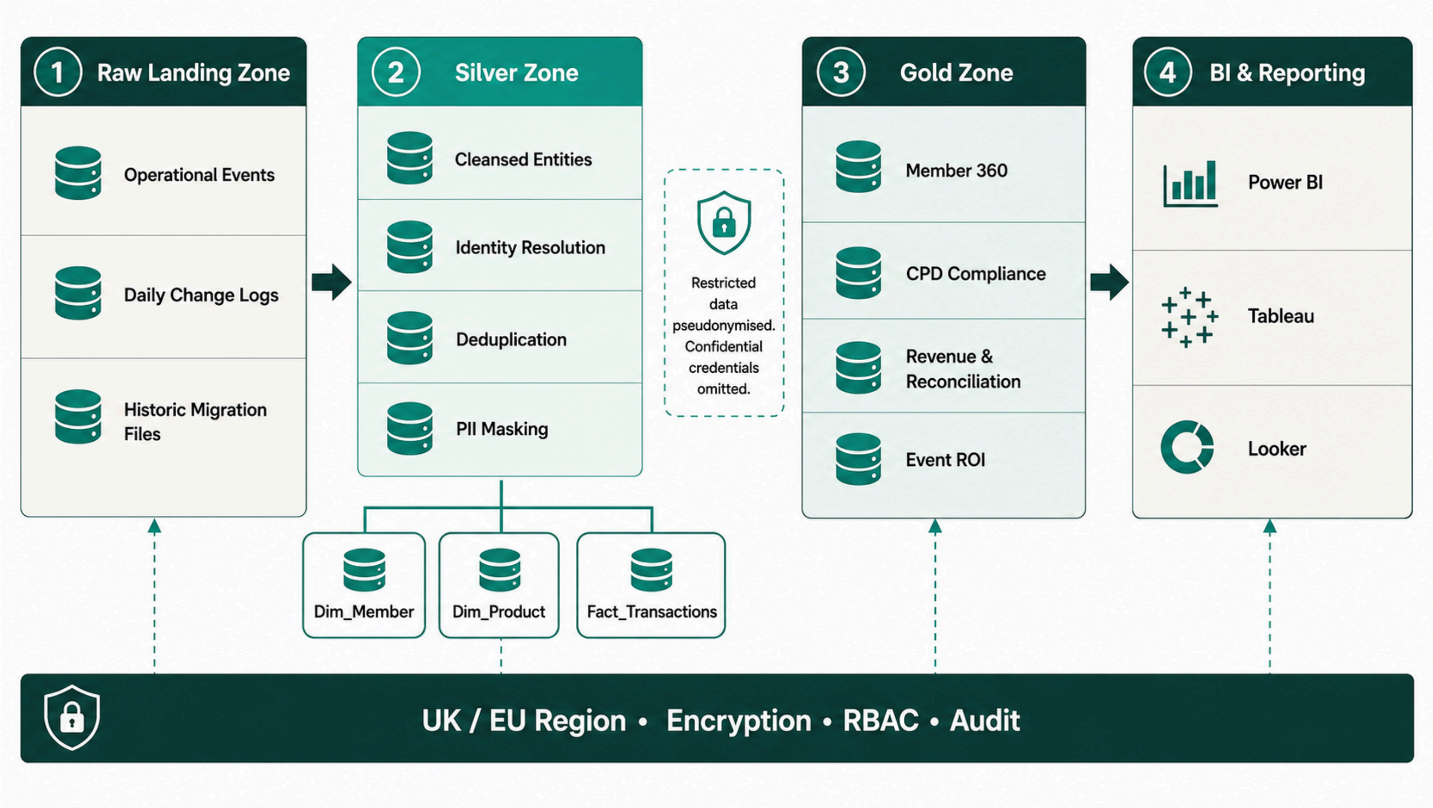


3.1 Platform Feature & Architecture Comparison

Evaluation Vector	Snowflake Data Cloud	Microsoft Fabric	Databricks Lakehouse
Core Architecture	Multi-cluster shared data warehouse with native Apache Iceberg lakehouse support.	Unified SaaS platform combining Data Factory, Synapse, and Power BI on OneLake.	Open lakehouse platform built on Apache Spark, Delta Lake, and Unity Catalog.
Storage Storage Format	Proprietary micro-partition format or Apache Iceberg (open standard).	Delta Lake (Parquet) natively in OneLake.	Delta Lake (Parquet) natively in cloud storage (S3/ADLS).
BI & Analytics Integration	Excellent connectivity with Power BI, Tableau, Looker via high-speed JDBC/ODBC.	**Native Direct Lake mode** for Power BI (queries OneLake without import refresh).	Databricks SQL Serverless with native Power BI & Tableau connectors.
Advanced Analytics / ML	Snowflake Cortex AI & Snowpark (Python/Java/Scala).	Integrated Azure ML & Fabric Data Science notebooks.	**Best-in-Class** MLflow, PySpark, custom AI/ML model training capabilities.
Data Governance & Security	Horizon Governance (RBAC, row/column masking, tag-based policy).	Microsoft Purview integration across Fabric & OneLake items.	**Unity Catalog** (centralized governance for files, tables, ML models across clouds).
Workato Ingestion Maturity	**Native Snowflake-Ready Connector** (Bulk stage loading, upserts, Snowpipe).	Native Microsoft Fabric & Azure Data Lake Storage Gen2 connectors.	Native Databricks Connector (supports SQL Warehouses & DBFS landing).
Management Overhead	**Near-Zero Operations** (Fully managed SaaS; auto-tuning, auto-scaling).	Low-to-Moderate (SaaS UI, but requires capacity	Moderate-to-High (Requires Spark cluster optimization

Evaluation Vector	Snowflake Data Cloud	Microsoft Fabric	Databricks Lakehouse
		allocation & workspace management).	and workspace management).

**Footnote: Cloud list prices change frequently — confirm against each vendor's pricing calculator before presenting final budget figures to the BSAVA steering committee.*

3.2 Pricing Models & Financial Breakdown

Understanding the pricing model of each vendor is essential to prevent cost overruns for BSAVA.

1. Snowflake Pricing Mechanics

- **Compute Billing**: Uses **Snowflake Credits**. Compute warehouses are billed per second with a **60-second minimum** charge upon startup.
  - Standard Warehouse (1 credit/hr ≈ 2.00–3.00 depending on region and edition).
  - Enterprise Edition adds multi-cluster warehouses and automatic scale-out.
  - Auto-suspend ensures warehouses shut down immediately when queries complete (e.g., set auto-suspend to 60 seconds).
- **Gen2 Standard Warehouses**: Snowflake Gen2 standard warehouses (GA Nov 2025) carry a ~1.25–1.35x credit multiplier over Gen1, but deliver up to 4.4x faster performance on DML-heavy workloads — highly relevant if BSAVA's migration cleansing workloads are DML-heavy.
- **Cortex AI Credits**: Snowflake introduced a separate **"AI Credits"** pricing currency (flat $2.00/credit, effective April 2026) for Cortex AI features, distinct from standard virtual warehouse compute credits.
- **Storage Billing**: Flat fee per Terabyte per month of compressed data (~$23/TB/month on AWS/Azure).
- **Iceberg Table Cost Advantage**: If BSAVA stores historical data in Apache Iceberg format on AWS S3/Azure ADLS, **Snowflake charges $0 for storage **—BSAVA only pays raw cloud storage costs ($ 18/TB/mo) and compute credits when querying.
- **Snowpipe Ingestion**: Charged at a simplified rate of **0.0037 credits per GB** ingested.

2. Microsoft Fabric Pricing Mechanics

- **Compute Billing**: Uses **Capacity Units (CUs)** packaged into **F SKUs** (e.g., F2 up to F2048).
  - Compute pool is shared across all Fabric workloads (Data Factory pipelines, Spark jobs, Data Warehouses, Power BI reports).
  - Billed hourly via Pay-As-You-Go (PAYG) or discounted via 1-year/3-year Reservation commitments.
  - *Example*: Published PAYG rates are ~$0.18/CU-hour. An F8 Capacity (8 CUs) costs approximately **$ 1,050–$1,150/month **on Pay-As-You-Go, or **$ 620–$680/month** under a 1-year reservation (reflecting the commonly cited ~41% 1-year commitment discount).
- **Storage Billing**: OneLake storage is billed separately at standard Azure Blob rates (~$0.023 per GB/month or $ 23/TB/month).
- **Direct Lake Cost Advantage**: Power BI reports read Delta tables directly from OneLake without requiring high-memory Import Mode dataset refreshes, saving significant BI server costs.

3. Databricks Pricing Mechanics

- **Retirement of Standard Tier**: Databricks retired its Standard tier on AWS and GCP in October 2025 (Azure retiring October 2026). Consequently, **Premium tier is now the effective baseline floor** for any new Databricks deployment, offering Unity Catalog governance, fine-grained access control, and enterprise security.
- **Two-Bill System**:

   i. **Databricks DBUs**: You pay Databricks per DBU consumed per second depending on workload tier (e.g., Premium Jobs Compute is ~$0.20/DBU, PremiumAll - PurposeComputeis$ 0.55/DBU, Serverless SQL is ~$0.70/DBU).
   ii. **Cloud Infrastructure Bill**: You pay AWS/Azure directly for the underlying Virtual Machines (EC2/Azure VMs) and Object Storage (S3/ADLS) hosting the Spark clusters.
- **Serverless SQL Option**: Databricks manages cloud infrastructure, simplifying billing into a single higher DBU rate.

3.3 Comparative Summary & TCO Matrix for BSAVA

Dimension	Snowflake	Microsoft Fabric	Databricks
Est. Monthly Cost (Baseline BSAVA Workload)	**Low to Moderate** (~$600–$1,200/mo)	**Low to Moderate** (~$620–$1,150/mo for F8)	**Moderate to High** (~$1,100–$2,000/mo Premium floor incl. Infra)
Predictability of Pricing	Highly predictable (Credits pause when idle; Gen2 DML efficiency; distinct Cortex AI credits).	Fixed capacity cost (F SKU capped per month; ~41% 1-yr reservation discount).	Variable (Premium tier floor; depends on Spark cluster runtimes & VM rates).
Best Suited For...	Standard SQL BI, cross-domain reporting, near-zero DBA admin effort.	Organizations committed to **Power BI** and the Microsoft Azure ecosystem.	Organizations with dedicated Data Engineering teams doing heavy ML/AI.
Ease of Workato Integration	**Extremely Easy** (Snowflake Bulk stage connector).	Easy (Fabric REST & ADLS connectors).	Moderate (JDBC / SQL Warehouse endpoint).

**Footnote: Cloud list prices change frequently — confirm against each vendor's pricing calculator before presenting final budget figures to the BSAVA steering committee.*

3.4 Strategic Recommendation for BSAVA

Tip

Primary Recommendation: Microsoft Fabric or Snowflake

1. **Option A — Microsoft Fabric (If BSAVA standardizes on Power BI)**:
   - If BSAVA utilizes Power BI for executive reporting, **Microsoft Fabric** is the most cost-effective and seamless platform. The **Direct Lake** technology allows Power BI to query OneLake Delta tables instantly without memory caching or nightly import refreshes.
2. **Option B — Snowflake Data Cloud (If BSAVA prioritizes simplicity & multi-BI tool flexibility)**:
   - If BSAVA wants a pure, low-maintenance SQL data warehouse that seamlessly integrates with Workato and any BI tool (Tableau, Power BI, Looker) with zero infrastructure management, **Snowflake** is the ideal solution. Using **Apache Iceberg tables** keeps long-term storage costs at raw cloud rates.

Databricks is deemed over-engineered for BSAVA's current scope unless advanced predictive data science or complex machine learning models become a core requirement.

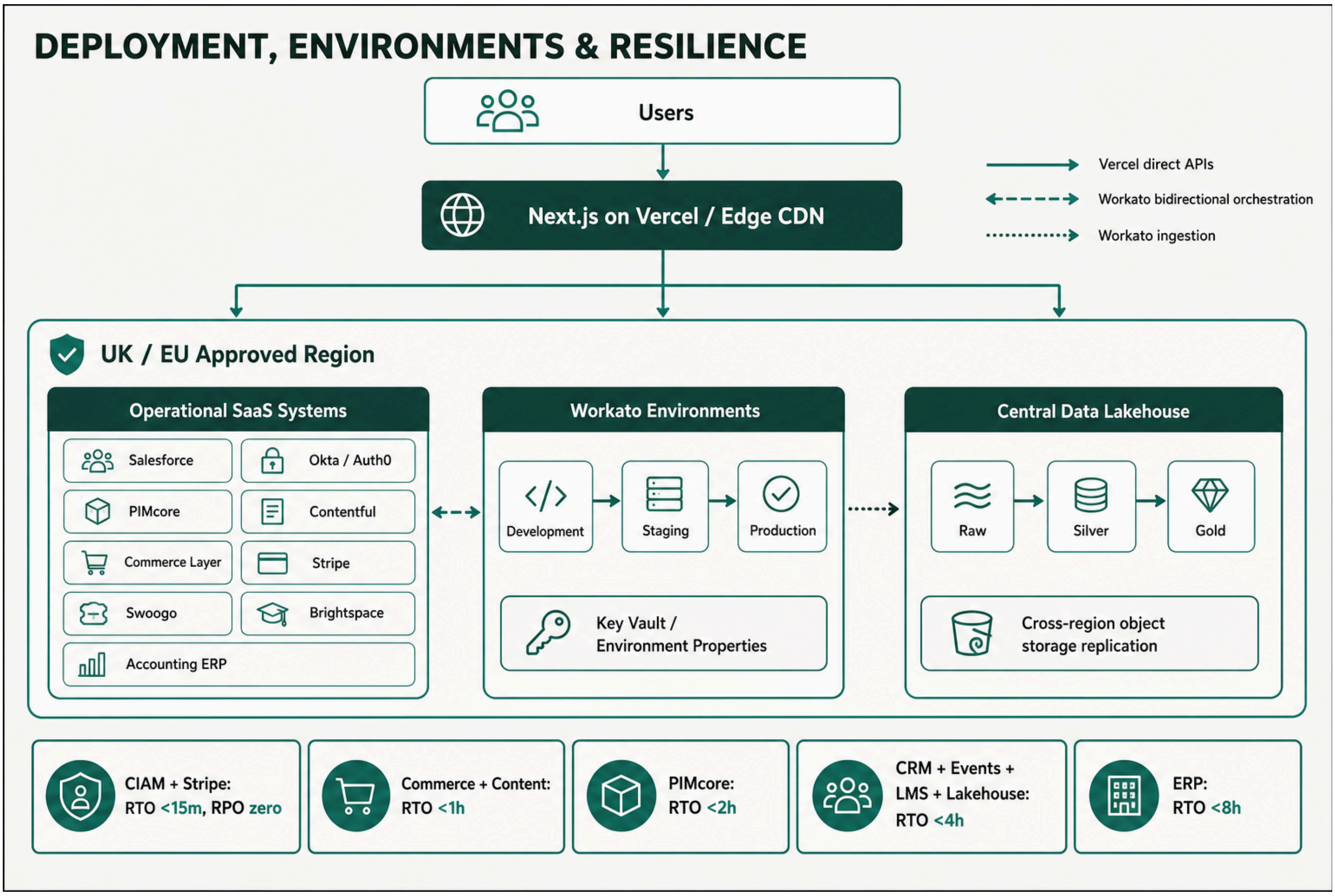
3.5 Cost Governance & FinOps Cadence

To ensure predictable cloud spending and prevent unbudgeted cost overruns across Workato and the Data Lakehouse, BSAVA will establish a formal monthly **FinOps & Consumption Review**:

1. **Workato Consumption Audits**:
   - Monthly review of recipe task consumption against BSAVA's annual Workato task allocation.
   - High-frequency webhooks will be audited to ensure payloads land in micro-batches rather than triggering redundant API executions.
2. **Lakehouse Compute & Capacity Audits**:
   - **Snowflake**: Enforce automatic warehouse auto-suspend set strictly to 60 seconds of inactivity. Perform monthly audits of warehouse sizing (e.g., restricting migration ETL jobs to Medium warehouses while daily BI queries run on X-Small warehouses).
   - **Microsoft Fabric**: Perform daily capacity smoothing checks in the Fabric Capacity Metrics App to identify burst spikes, background queue delays, or throttling risks on the F8 SKU.
   - **Databricks**: Monitor cluster auto-scaling limits and enforce maximum run-times on interactive All-Purpose clusters.
3. **Automated Alerting Thresholds**:
   - Configure cloud budget alerts at 50%, 80%, and 100% of the allocated monthly lakehouse compute budget.
   - Automated Slack notifications sent to `#platform-ops` when monthly consumption exceeds expected run-rates by > 15%.

3.6 Disaster Recovery & RTO/RPO Targets

To guarantee operational resilience and compliance, the following **Recovery Time Objective (RTO)** and **Recovery Point Objective (RPO)** targets are defined for all Systems of Record and the Central Data Lakehouse:



System / Platform	Role	Disaster Recovery Strategy	Target RTO	Target RPO
Salesforce (CRM)	Customer & Contact History SoR	Native multi-region failover & daily automated metadata/data backups	< 4 Hours	< 1 Hour
Okta / Auth0 (CIAM)	Identity & Authentication	High-availability multi-tenant cloud mesh with cross-region redundancy	< 15 Minutes	Zero (< 1 Sec)
PIMcore (PIM / DAM)	Product Catalog, Asset & Entitlement SoR	Multi-AZ database cluster with automated snapshot replication	< 2 Hours	< 15 Minutes
Contentful (CMS)	Editorial Content Master	Global Edge CDN distribution with multi-region platform failover	< 1 Hour	< 15 Minutes
Commerce Layer (OMS)	Cart, Order & Membership Tier SoR	Multi-region serverless deployment with automated database backups	< 1 Hour	< 5 Minutes
Stripe	Payment Processing	Globally distributed PCI-DSS Level 1 infrastructure	< 15 Minutes	Zero (< 1 Sec)
Swoogo	Event & Delegate Management	SaaS cloud redundancy with continuous snapshot backup	< 4 Hours	< 1 Hour
Brightspace (LMS)	Learning & CPD Accreditation	Cloud hosted multi-zone deployment with daily Data Hub extracts	< 4 Hours	< 1 Hour
Accounting (ERP)	Financial Ledger	Scheduled cloud backup with offsite cold storage archiving	< 8 Hours	< 24 Hours
Central Data Lakehouse	Enterprise Analytics & BI	Cloud object storage cross-region replication (S3/ADLS) & time-travel point-in-time recovery	< 4 Hours	< 1 Hour

4. Historic Data Migration Strategy (Phased Plan)

Migrating legacy historical data (from BSAVA's legacy CRM, membership databases, and historical transaction logs) into the modern composable architecture presents a major challenge:

- **Operational Need**: Active contacts, open orders, current product catalogs, and active course enrolments must be loaded into their respective operational SoRs (Salesforce, PIMcore, Swoogo, Brightspace).
- **Analytical Need**: Multi-year historical reporting, past financial audits, inactive member histories, and legacy CPD hours must be preserved *without bloating operational SoRs*.

To solve this, BSAVA adopts a **Dual-Path Historic Migration Strategy**:

```
                    ┌──────────────────────────────────────────┐
                    │      LEGACY SOURCE DATA REPOSITORY       │
                    │  (Legacy CRM, SQL DBs, Historic CSVs)    │
                    └─────────────────────┬────────────────────┘
                                          │
                   Phase 1: Discovery &   │ Profiling
                                          ▼
                    ┌──────────────────────────────────────────┐
                    │     WORKATO MIGRATION STAGING BUCKET     │
                    └────────────┬─────────────────┬───────────┘
                                 │                 │
  PATH A: Operational Data Load  │                 │   PATH B: Full Historical Archive Load
  (Active Records & Entities Only)│                │   (100% Raw Legacy History)
                                 ▼                 ▼
            ┌────────────────────────────┐ ┌────────────────────────────┐
            │ Operational SoR Ingestion  │ │ Data Lakehouse Raw Zone    │
            │ - Salesforce (Active Mbrs) │ │ - raw_legacy_contacts      │
            │ - PIMcore (Active SKUs)    │ │ - raw_legacy_orders        │
            │ - Brightspace (Active Crs) │ │ - raw_legacy_cpd_hours     │
            └─────────────┬──────────────┘ └──────────────┬─────────────┘
                          │                               │
                          │   Phase 4: Reconciliation     │
                          └───────────────┬───────────────┘
                                          ▼
                    ┌──────────────────────────────────────────┐
                    │      UNIFIED MEMBER 360 & BI DATAMART    │
                    │  (Joined Operational + Historic Delta)   │
                    └──────────────────────────────────────────┘
```

4.1 Phased Execution Roadmap

Phase 1: Discovery, Data Quality Profiling & Schema Mapping (Weeks 1–3)

- **Tasks**:
  i. Profile legacy databases to identify active vs. inactive contact records, duplicate email addresses, corrupted transaction logs, and orphaned membership records.
  ii. Produce an **Entity Field Mapping Document** mapping legacy table fields to destination SoR schemas (e.g., Legacy `MEMB_MAST` → Salesforce `Contact` + Commerce Layer `Subscription` + PIMcore `Entitlement`).
  iii. Define deduplication and identity resolution rules (Primary Key: Verified Email Address).

Phase 2: Raw Historic Snapshot to Data Lakehouse (Weeks 4–5)

- **Tasks**:
  i. Extract 100% of raw legacy tables into Parquet/JSON files.
  ii. Load raw files directly into the **Central Data Lakehouse** (`raw_legacy_` schemas).
  iii. *Outcome*: Guarantees that historical reporting continuity (e.g., 10-year membership trends, historic financial reporting) is preserved immediately, even for contacts who will not be migrated into Salesforce.

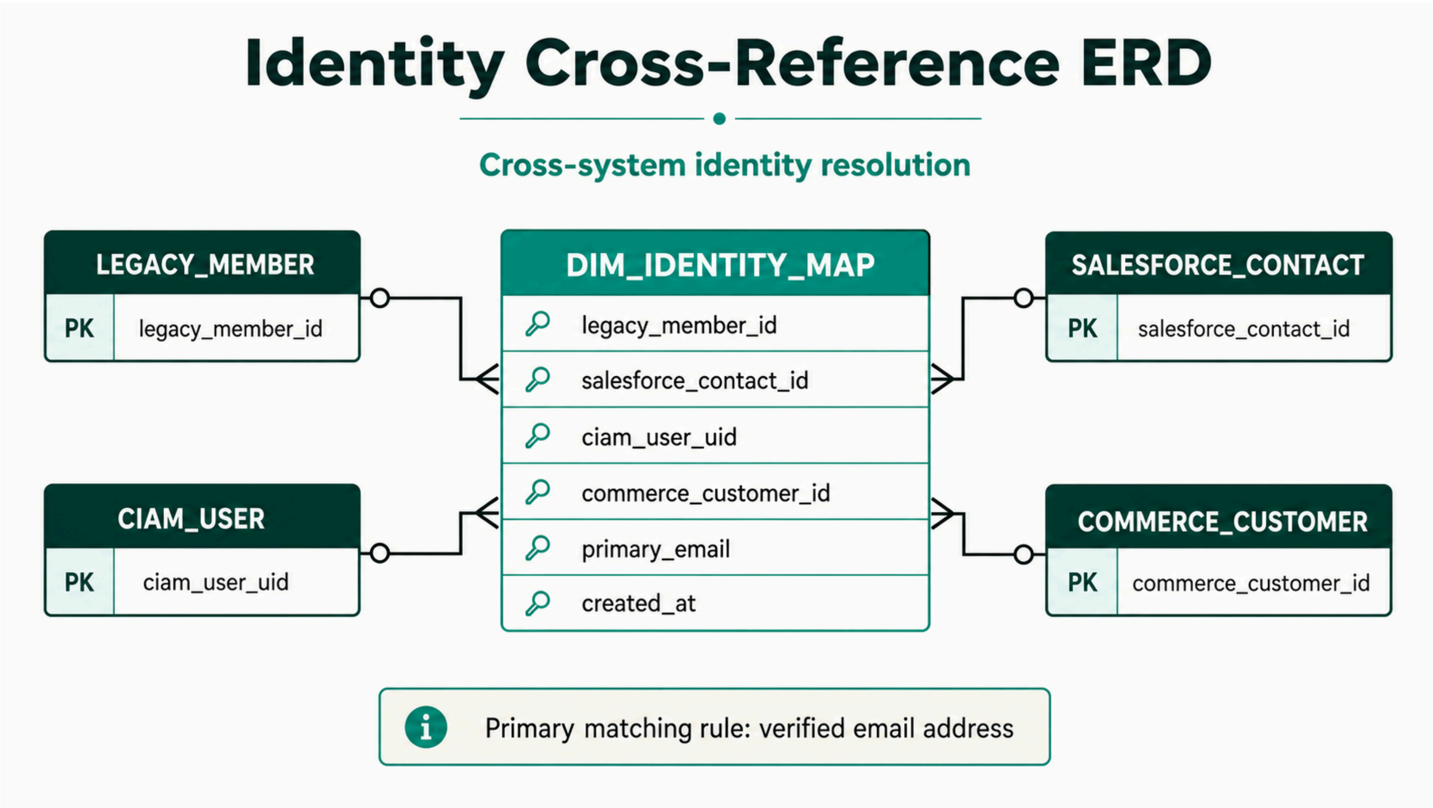
Phase 3: Operational Cleansing, Transformation & SoR Population (Weeks 6–9)

- **Tasks**:
  i. Filter legacy dataset to extract only **Active Entities**:
     - Contacts with active memberships or transactions within the last 36 months.
     - Active product SKUs, current event schedules, and active course catalogs.

ii. Execute Workato migration recipes to populate target operational SoRs in sequence:
   - **Step 3.1**: Load PIMcore with Product Catalog, Assets, and Entitlement Access Rules.
   - **Step 3.2**: Load Salesforce with Member Accounts, Contacts, and Committee Roles.
   - **Step 3.3**: Load Commerce Layer with Active Membership Subscriptions & Tiers.
   - **Step 3.4**: Load Brightspace with Student Profiles & Course History.
   - **Step 3.5**: Load Swoogo with Upcoming Event Registrations.

iii. Log cross-system mapping keys during insertion (e.g., mapping `Legacy_ID_123` to `Salesforce_Contact_ID_003XXX`).

Phase 4: Cross-System Identity Resolution & Cross-Reference Mapping (Weeks 10–11)



- **Tasks**:

i. Populate a centralized `Identity_Cross_Reference` table in the Data Lakehouse:

```
CREATE TABLE diember
    legacy_member_id    VARCHAR,
    salesforce_contact_id VARCHAR,
    ciAM_user_uid       VARCHAR,
    commerce_customer_id VARCHAR,
    primary_email       VARCHAR,
    created_at          TIMESTAMP
);
```

ii. Execute automated reconciliation scripts to verify that every active member in Salesforce maps to an identity in CIAM (Okta) and historical order records in the Data Lakehouse.

Phase 5: Parallel Run, Automated Audit & Final Cutover (Weeks 12–13)

- **Tasks**:

i. Execute parallel reporting runs comparing legacy report outputs against the Data Lakehouse BI dashboards.

ii. Validate financial balances: Total legacy revenue matches Commerce Layer/Stripe + Historical Data Lake total to within £0.00 variance.

iii. Perform cutover: Decommission legacy write access, enable live Workato production recipes, and transition business teams to the new MACH platform.

- **Rollback Criteria & Contingency Plan for Phase 5 Cutover**:
  - **Reconciliation Failure Trigger**: If the mandatory £0.00 financial reconciliation variance between legacy systems and Commerce Layer / Stripe / Data Lakehouse is **not achieved on the first attempt** (or if entitlement discrepancy exceeds 0.01%), cutover execution is immediately halted.
  - **Parallel Run Extension**: The cutover date is formally deferred, and the parallel run window is extended by **2 to 4 weeks** while data engineering teams reconcile variances.
  - **Operational Fallback Protocol**: Legacy systems remain in active read-write mode. Live Workato production recipes will be switched back to passive read-only extraction mode to prevent half-committed operational state in modern SoRs.
  - **Escalation Path**: Incident report and reconciliation delta analysis presented to the IT Steering Committee within 24 hours for sign-off on remediation steps before scheduling a re-cutover attempt.

4.2 Data Cleansing & Migration Governance Rules

1. **Deduplication Rule**:
   - If legacy records contain multiple contacts sharing the same verified email address, merge attributes into the most recently active record. Retain legacy IDs in an array attribute (`legacy_ids_merged`) in Salesforce and Data Lakehouse.
2. **Inactive Member Archival Rule**:
   - Members with no activity for > 5 years are **not** migrated into Salesforce (saving Salesforce license and storage costs). They reside exclusively in the Data Lakehouse raw zone to comply with retention policies while remaining available for historical reporting.
3. **CPD Credit Preservation**:
   - Historical CPD hours are written directly to Salesforce Contact summary fields (`Historic_CPD_Hours__c`) and detailed in the Data Lakehouse `fact_cpd_history` table for complete member transcripts.

5. Client Workshop Agenda & Actionable Next Steps

To finalize decisions with BSAVA leadership and technical stakeholders, we propose the following 1-day executive workshop agenda:

Proposed Workshop Agenda

Session Topic	Objective / Key Decisions Required
Information Architecture & SoR Approval	Review Data Ownership Matrix. Formally sign off on Salesforce, PIMcore, Swoogo, and Brightspace SoR boundaries.
Workato Data Orchestration Architecture	Review operational vs. analytical recipes, token management & cost control, error handling (DLQ), and Workato environment setup.
Data Lakehouse Platform Selection	Evaluate **Snowflake vs. Microsoft Fabric**. Review BI integration strategy (Power BI vs. alternative) and approve cloud budget.
Historic Data Migration Strategy	Review 5-Phase Roadmap, active member filtering thresholds, and financial reconciliation criteria.
Roadmap, Milestones & Action Items	Confirm project team roles, sign off on migration timelines, and establish weekly governance steering calls.

Actionable Next Steps Prior to Workshop:

1. **BSAVA IT Team**: Provide sample data extracts (anonymized CSVs) of legacy CRM contact and transaction tables for migration profiling.
2. **BSAVA BI Team**: Confirm preferred enterprise BI visualization tool (Power BI vs. Tableau) to finalize the Data Lakehouse recommendation (Fabric vs. Snowflake).
3. **Timberyard Architecture Team**: Finalize Workato recipe staging templates for Phase 2 raw historic data landing.

End of Strategy Document.